

HealthMatch Hub - Project Report

Abstract

HealthMatch Hub is a MERN Stack web application that allows users to search doctors and hospitals, book appointments, and manage healthcare services online.

Objectives

Develop a secure healthcare appointment system with authentication, doctor and hospital management, appointment scheduling, and an admin dashboard.

Technology Stack

Frontend: React.js, Vite, Bootstrap.

Backend: Node.js, Express.js.

Database: MongoDB.

Authentication: JWT.

Tools: Git, GitHub.

Modules

User Authentication

Doctor Management

Hospital Management

Appointment Booking

My Appointments

Admin Dashboard

Features

JWT Authentication, Password Hashing, Role-Based Access Control, CRUD Operations, Responsive UI, Doctor Search, Appointment Booking.

Database Design

Collections: Users, Doctors, Hospitals, Appointments.

Workflow

User Login → Browse Doctors/Hospitals → Book Appointment → Manage Appointments → Admin Controls.

Future Scope

Online Payments, Email/SMS Notifications, Video Consultation, AI-based Recommendations, Medical Record Management.

Conclusion

The project successfully demonstrates a complete MERN-based healthcare management platform with secure authentication and scalable architecture.